Chapter 10. Linear Regression

Least squares, permutation and bootstrap inference, robust regression and M-estimation

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10.1 Simple linear regression

Motivation. Least squares is sensitive to outliers in y. Rank-based slope estimators, such as the median of pairwise slopes, give robust alternatives with distribution-free confidence intervals.

Model $y_i = \alpha + \beta x_i + \epsilon_i$ with uncorrelated errors of mean 0 and variance σ^2; inference is conditional on the x_i. Least squares gives $\hat\beta = S_{xy}/S_{xx}$, $\hat\alpha = \bar y - \hat\beta \bar x$ and $s^2 = \sum e_i^2/(n-2)$.

Proposition 10.1

$E\hat\beta = \beta$ and $\operatorname{Var}\hat\beta = \sigma^2/S_{xx}$. The residuals satisfy $\operatorname{Var}(e_i) = \sigma^2(1-h_i)$ with leverage $h_i = \frac{1}{n} + \frac{(x_i-\bar x)^2}{S_{xx}}$, so the adjusted residuals $r_i = e_i/\sqrt{1-h_i}$ have variance σ^2.

Proof. $\hat\beta = \sum w_i y_i$ with $w_i = (x_i - \bar x)/S_{xx}$, $\sum w_i = 0$, $\sum w_i x_i = 1$, $\sum w_i^2 = 1/S_{xx}$. With the hat matrix H, $\mathbf{e} = (I-H)\boldsymbol\epsilon$ and $\operatorname{Var}\mathbf{e} = \sigma^2(I-H)$, whose diagonal entries are $\sigma^2(1-h_i)$. ■

Under normal errors, $(\hat\beta - \beta)/\widehat{\mathrm{se}}(\hat\beta) \sim t_{n-2}$ with $\widehat{\mathrm{se}}(\hat\beta) = s/\sqrt{S_{xx}}$. Without normality we use resampling.

Permutation test of $H_0 : \beta = 0$. Under H_0, $y_i = \alpha + \epsilon_i$ are exchangeable, so shuffle the y's against the fixed x's. The statistics $\hat\beta$, $\sum x_i y_i$ and $t = \hat\beta/\widehat{\mathrm{se}}(\hat\beta)$ give the same test: with S_{xx} and S_{yy} fixed, $t = \hat\beta\sqrt{(n-2)S_{xx}}/\sqrt{S_{yy} - \hat\beta^2 S_{xx}}$ is increasing in $\hat\beta$. This is the same test as the permutation test of correlation (Section 5.1).

Bootstrap intervals. Use the case bootstrap (random x, robust to model failure) or the residual bootstrap $y_i^* = \hat\alpha + \hat\beta x_i + r_i^*$ (fixed design, model assumed). Because $\widehat{\mathrm{se}}(\hat\beta)$ has a formula, the bootstrap-t interval

$$\beta \in \big(\hat\beta - \widehat{\mathrm{se}}(\hat\beta) Q_{t^*}(1-\tfrac{\alpha}{2}),\ \hat\beta - \widehat{\mathrm{se}}(\hat\beta) Q_{t^*}(\tfrac{\alpha}{2})\big), \qquad t^* = \frac{\hat\beta^* - \hat\beta}{\widehat{\mathrm{se}}(\hat\beta^*)},$$

is cheap and second-order accurate. For tests of $\beta = 0$:

- null resampling: resample x and y *independently* (case form), or $y^* = \bar y + r_0^*$ with $r_{0,i} = (y_i - \bar y)/\sqrt{1 - 1/n}$ (residual form);
- pivot resampling: use t^* above and compare with $t_{\mathrm{obs}} = \hat\beta/\widehat{\mathrm{se}}(\hat\beta)$, which inverts the bootstrap-t interval.

10.2 Multiple regression

$\mathbf{y} = X\boldsymbol\beta + \boldsymbol\epsilon$ with X of size $n \times (p+1)$: $\hat{\boldsymbol\beta} = (X'X)^{-1}X'\mathbf{y}$, $E\hat{\boldsymbol\beta} = \boldsymbol\beta$, $\operatorname{Var}\hat{\boldsymbol\beta} = \sigma^2(X'X)^{-1}$, $s^2 = \mathbf{e}'\mathbf{e}/(n-p-1)$ (unbiased since $E\,\boldsymbol\epsilon'(I-H)\boldsymbol\epsilon = \sigma^2\operatorname{tr}(I-H)$). Each β_j is a *partial* effect: the effect of x_j adjusted for the other regressors.

Comparing nested models. Split $X = [X_0, X_1]$ ($k+1$ and $q = p-k$ columns) and test $H_0 : \boldsymbol\beta_1 = 0$. With hat matrices H and H_0,

$$F = \frac{\mathbf{y}'(H - H_0)\mathbf{y}/q}{\mathbf{y}'(I-H)\mathbf{y}/(n-p-1)} = \frac{\hat{\boldsymbol\beta}_1'\{X_1'(I-H_0)X_1\}\hat{\boldsymbol\beta}_1}{q\,s^2}. \tag{10.1}$$

Proposition 10.2

Under H_0 with normal errors, $F \sim F_{q,n-p-1}$. More generally $F_{\boldsymbol{\beta}_1}$, defined by replacing $\hat{\boldsymbol{\beta}}_1$ with $\hat{\boldsymbol{\beta}}_1 - \boldsymbol{\beta}_1$ in (10.1), has this distribution for every $\boldsymbol{\beta}_1$, so it is a pivot.

Proof. $H - H_0$ is a projection of rank q orthogonal to $I - H$, and both annihilate X_0. Under H_0, the numerator and denominator quadratic forms reduce to $\boldsymbol{\epsilon}'(H - H_0)\boldsymbol{\epsilon}$ and $\boldsymbol{\epsilon}'(I - H)\boldsymbol{\epsilon}$, independent $\sigma^2\chi^2_q$ and $\sigma^2\chi^2_{n-p-1}$ (Cochran). The second form of (10.1) is Frisch–Waugh–Lovell: $\hat{\boldsymbol{\beta}}_1$ is the regression of $(I - H_0)\mathbf{y}$ on $\tilde{X}_1 = (I - H_0)X_1$, and $H - H_0$ projects onto the column space of $\tilde{X}_1$. For general $\boldsymbol{\beta}_1$, apply the same argument to $\mathbf{y} - X_1\boldsymbol{\beta}_1$. ■

For $q = 1$, $F = t_j^2$, so the F-test of one coefficient is the two-sided t-test. A one-sided alternative $\beta_j > 0$ needs t_j itself, and the permutation and bootstrap schemes below apply to t_j in the same way. For bootstrap tests in multiple regression, null resampling uses either the case form (draw the $\mathbf{x}$-rows and the y's independently, which is appropriate when all slopes are tested) or residuals of the reduced model with replacement. Pivot resampling uses the case or full-model residual bootstrap with $F_{\hat{\boldsymbol{\beta}}_1}$ or $t_j^* = (\hat{\beta}_j^* - \hat{\beta}_j)/\widehat{\mathrm{se}}(\hat{\beta}_j^*)$, and the p-value is $\#\{F^{*b} \geq F_{\mathrm{obs}}\}/(B+1)$ (with the observed value counted).

Permutation tests for $H_0 : \boldsymbol{\beta}_1 = 0$. Under H_0 the errors are exchangeable but the y_i are not (their means depend on X_0), so no permutation test is exact. Three approximate schemes:

1. *Shuffle y:* effectively tests $\beta_1 = \cdots = \beta_p = 0$; it can be anticonservative and is not recommended.
2. *Freedman–Lane:* fit the reduced model, shuffle its residuals e_0, set $y^* = \hat{y}_0 + e_0^*$, and recompute F. This is generally the most accurate.
3. *ter Braak:* shuffle the full-model residuals, set $y^* = \hat{y} + e^*$, and recompute the pivot $F_{\hat{\boldsymbol{\beta}}_1}$.

All three are asymptotically equivalent. Replacing "shuffle" with "resample with replacement" in schemes 2 and 3 gives the residual bootstrap tests under the null and with the pivot, respectively. Bootstrap-t intervals for each β_j use $t_j^* = (\hat{\beta}_j^* - \hat{\beta}_j)/\widehat{\mathrm{se}}(\hat{\beta}_j^*)$ with $\widehat{\mathrm{se}}^2(\hat{\beta}_j) = s^2(X'X)^{-1}_{jj}$.

10.3 Robust regression

Least squares is the maximum likelihood estimator under normal errors. It loses efficiency under heavy-tailed errors and can be ruined by a single outlier.

Definition (Breakdown point)

The smallest fraction of observations that, sent to infinity, can send the estimator to infinity. Roughly: the mean has $1/n \to 0$, the α-trimmed mean α, and the median $1/2$.

Trimmed and Winsorized means. The α-trimmed mean drops the lowest and highest α fractions of the data and averages the rest. The α-Winsorized mean instead replaces them by the α and $1 - \alpha$ sample quantiles and then averages all n values. Both have breakdown point α. The Hodges–Lehmann estimator has breakdown point $1 - 1/\sqrt{2} \approx 29\%$: sending a fraction ε of points to infinity drags a fraction $1 - (1 - \varepsilon)^2$ of the Walsh averages with them, which exceeds $\frac{1}{2}$ when $\varepsilon > 1 - 1/\sqrt{2}$.

Example 10.1

For $-2, -1, -1, 0, 0, 1, 1, 2, 3, 50$: mean 5.3; 10% trimmed mean 0.625 (drop -2 and 50); 10% Winsorized mean 0.7 (replace them by -1 and 3); median 0.5. Replacing 50 by 10^{100} changes only the mean.

Trimming and Winsorizing treat both tails alike, even when the contamination is one-sided. M-estimators instead down-weight observations according to their own residuals.

10.3.1 M-estimators

For location, the least squares estimator $\bar{x} = \arg\min_\mu \sum (x_i - \mu)^2$ is generalized to

$$\hat{\mu}^\Psi = \arg\min_\mu \sum_i \Psi\Big(\frac{x_i - \mu}{\hat{\tau}}\Big) \iff \sum_i \psi\Big(\frac{x_i - \mu}{\hat{\tau}}\Big) = 0, \qquad \psi = \Psi',$$

with a robust scale $\hat{\tau} = 1.4826$ MAD (consistent for σ under normality, since $\Phi^{-1}(3/4) = 0.6745$).

- $\psi(x) = x$: the mean. $\psi(x) = \operatorname{sgn}(x)$: the median.
- **Huber:** $\psi(x) = \max(-k, \min(k, x))$, quadratic Ψ in the center and linear in the tails. With $k \approx 1.35$ it has about 95% efficiency at the normal.
- **Tukey biweight:** $\psi(x) = x(1 - (x/c)^2)^2 I(|x| \le c)$, redescending, so gross outliers get zero weight.

A bounded ψ with a 50% breakdown scale gives breakdown point 50%. ("M" stands for maximum-likelihood type: with $\Psi(x) = x^2/2$ and $\hat{\tau} = \sigma$, $\hat{\mu}^\Psi$ is the normal maximum likelihood estimator.) The equation has no closed form and is solved numerically, e.g. by iteratively reweighted least squares with weights $\psi(u_i)/u_i$.

Inference on a location with outliers. With a contaminated sample, the t-test and t-interval are driven by the outliers: the mean and s both inflate. Robust alternatives:

- the sign test and order-statistic interval (Chapter 2), valid for the median under any continuous F;
- the signed-rank test and Walsh-average interval, valid if the uncontaminated distribution is symmetric;
- an M-estimator with a bootstrap standard error and bias, giving a normal-approximation bootstrap interval, or better a percentile or BCa interval. Its asymptotic standard error requires the influence function, so the bootstrap is the practical route.

The lower the breakdown point of the underlying estimator, the wider the resulting interval tends to be under contamination. In regression, $\hat{\boldsymbol{\beta}}^\Psi = \arg\min \sum \Psi((y_i - \mathbf{x}_i'\boldsymbol{\beta})/\hat{\tau})$. The estimating equations $\sum \psi(\cdot)\,\mathbf{x}_i = 0$ show that Huber's estimator is robust to outlying y but not to outlying x (high leverage), because the factor x_{ij} is unbounded. **Yohai's MM-estimator** uses a redescending ψ (biweight, $c = 4.685$ for 95% efficiency) started from a high-breakdown S-estimate of scale, achieving a 50% breakdown point together with high efficiency. **Least trimmed squares** minimizes the sum of the h smallest squared residuals; $h = \lfloor n/2 \rfloor + \lfloor (p+1)/2 \rfloor$ gives maximal breakdown, and $h = \lfloor 0.9n \rfloor$ a 10% trimming. **Least median of squares** minimizes the median squared residual. Both are very robust but inefficient, and are mainly used as starting values.

10.3.2 The Theil–Sen estimator

For $y_i = \alpha + \beta x_i + \epsilon_i$, form the pairwise slopes $\beta_{ij} = (y_j - y_i)/(x_j - x_i)$ for $x_i \neq x_j$ and set

$$\hat{\beta}^{TS} = \text{median}_{i<j}\, \beta_{ij}, \qquad \hat{\alpha}^{TS} = \text{median}_i(y_i - \hat{\beta}^{TS} x_i).$$

Proposition 10.3 (Link to Kendall's tau)

Let $\tau_n(\beta_0)$ be Kendall's statistic between x_i and $y_i - \beta_0 x_i$. Then $\binom{n}{2}\tau_n(\beta_0) = \#\{\beta_{ij} > \beta_0\} - \#\{\beta_{ij} < \beta_0\}$. Hence the test of $H_0 : \beta = \beta_0$ by Kendall's tau, and the interval $(\beta_{(l)}, \beta_{(u)})$ with $l = \binom{n}{2}\frac{Q_K(\alpha/2)+1}{2}$, $u = \binom{n}{2}\frac{Q_K(1-\alpha/2)+1}{2}$ (Q_K the null quantile of τ_n), are exact. $\hat{\beta}^{TS}$ is the corresponding point estimate.

Proof. $\text{sgn}[(x_i - x_j)\{(y_i - \beta_0 x_i) - (y_j - \beta_0 x_j)\}] = \text{sgn}[(x_i - x_j)^2(\beta_{ij} - \beta_0)] = \text{sgn}(\beta_{ij} - \beta_0)$. At the true β, $y_i - \beta x_i = \alpha + \epsilon_i$ is independent of x_i, so $\tau_n(\beta)$ has the null law K_n. The acceptance region $Q_K(\frac{\alpha}{2}) \le \tau_n(\beta_0) \le Q_K(1 - \frac{\alpha}{2})$ is a range of $\#\{\beta_{ij} > \beta_0\}$, which is a step function of β_0 dropping by one at each ordered slope. So the region is an interval of ordered slopes with the stated indices. ■

$\hat{\beta}^{TS}$ has breakdown point $1 - 1/\sqrt{2} \approx 29\%$ and about 96% ARE relative to least squares under normal errors.

Example 10.2

Points $(1,1), (2,2), (3,3.5), (4,4), (5,20)$, where the last point is an outlier. The ten ordered pairwise slopes are $0.5, 1, 1, 1, 1.25, 1.5, 4.75, 6, 8.25, 16$, so $\hat{\beta}^{TS} = 1.375$ and $\hat{\alpha}^{TS} = -0.625$. Least squares gives slope 4.0, pulled up by a single point.

In multiple regression Theil–Sen has no unique definition. One version uses **backfitting**:

1. Start with single-variable Theil–Sen slopes $\hat{\beta}_j^{(0)}$ and set $\hat{\beta}_0^{(0)} = \text{median}_i(y_i - \sum_j \hat{\beta}_j^{(0)} x_{ij})$.
2. For $k = 1, 2, \ldots$ and each j, compute the Theil–Sen slope of the partial responses $y_i - \sum_{l \neq j} \hat{\beta}_l^{(k-1)} x_{il}$ on x_{ij}, then update the intercept as a median.
3. Iterate until the coefficients stabilize.

Inference with robust regression estimators. M- and MM-estimates come with approximate standard errors from the final weighted least squares step. These are reasonable for large n, and the bootstrap standard errors are usually somewhat larger and more reliable. With an approximate standard error one can form a bootstrap-t interval $(\hat{\beta}_j - \widehat{\text{se}}\, Q_{t^*}(1 - \frac{\alpha}{2}), \hat{\beta}_j - \widehat{\text{se}}\, Q_{t^*}(\frac{\alpha}{2}))$ from case resampling, or permutation tests of $t_j = \hat{\beta}_j / \widehat{\text{se}}(\hat{\beta}_j)$. Shuffling y alone tests the joint null that all slopes vanish; for a single coefficient use a residual-shuffling scheme as in Section 10.2. Bootstrap samples contain many repeated points, which can upset estimators based on pairwise slopes (ties in x); permutation tests avoid this.

10.4 Selected exercises

Motivation. The exercises give an exact permutation test for any hypothesized slope, prove the Frisch–Waugh–Lovell identity used for nested models, and establish the basic properties of the Huber location estimator.

Exercise 10.1 (Permutation test of $\beta = \beta_0$)

In $y_i = \alpha + \beta x_i + \epsilon_i$ with fixed x_i and i.i.d. errors, describe a permutation test of $H_0 : \beta = \beta_0$ against $\beta \neq \beta_0$ for a given $\beta_0 \neq 0$, and show that it is exact.

Solution.

Step 1. (Why not shuffle y.) Under H_0, $Ey_i = \alpha + \beta_0 x_i$ depends on i, so the y_i are not exchangeable and shuffling them is not justified.

Step 2. (Transform.) Put $z_i = y_i - \beta_0 x_i = \alpha + (\beta - \beta_0)x_i + \epsilon_i$. Under H_0, $z_i = \alpha + \epsilon_i$ are i.i.d., hence exchangeable, and unrelated to the fixed x_i.

Step 3. (Test.) Use the least squares slope of z on x, $b(z) = S_{xz}/S_{xx} = \hat{\beta} - \beta_0$, or its t-statistic. Recompute it for the $n!$ (or M random) permutations of z against the fixed x, and take $p = \#\{\pi : |b(z_\pi)| \geq |b(z)|\}/n!$.

Step 4. (Exactness.) The vector $(z_1, \ldots, z_n)$ is exchangeable under H_0 and the permutations form a group, so Theorem 3.3 gives $P_{H_0}(p \leq \alpha) \leq \alpha$, with equality when $\alpha\, n!$ is an integer and there are no ties. With M random permutations, Proposition 3.4 applies.

Collecting the β_0 that are not rejected gives an exact confidence set for β (Proposition 1.2). ■

Exercise 10.4 (Partial regression (Frisch–Waugh–Lovell))

Let $X = [X_0, X_1]$ have full column rank $p + 1$, with q columns in X_1. Let $M_0 = I - H_0$, $\tilde{X}_1 = M_0 X_1$ (each column of X_1 replaced by its residuals on X_0) and $\mathbf{y}_{|0} = M_0 \mathbf{y}$. Consider the partial regression $\mathbf{y}_{|0} = \tilde{X}_1 \boldsymbol{\gamma} + \text{error}$. Show:
(a) its least squares estimator is $\hat{\boldsymbol{\gamma}} = \hat{\boldsymbol{\beta}}_1$, and its residual vector equals the full-model residual vector $\mathbf{e}$;
(b) the F-statistic for $\boldsymbol{\gamma} = 0$, computed with $n - p - 1$ residual degrees of freedom, equals (10.1).

Solution.

(a)

Step 1. The full fit is $\mathbf{y} = X_0 \hat{\boldsymbol{\beta}}_0 + X_1 \hat{\boldsymbol{\beta}}_1 + \mathbf{e}$ with $X_0' \mathbf{e} = 0$ and $X_1' \mathbf{e} = 0$ (normal equations).

Step 2. Apply M_0. Since $M_0 X_0 = 0$, and $H_0 \mathbf{e} = X_0 (X_0' X_0)^{-1} X_0' \mathbf{e} = 0$ gives $M_0 \mathbf{e} = \mathbf{e}$,

$$\mathbf{y}_{|0} = \tilde{X}_1 \hat{\boldsymbol{\beta}}_1 + \mathbf{e}.$$

Step 3. $\tilde{X}_1' \mathbf{e} = X_1' M_0 \mathbf{e} = X_1' \mathbf{e} = 0$. So $\boldsymbol{\gamma} = \hat{\boldsymbol{\beta}}_1$ solves the normal equations $\tilde{X}_1' (\mathbf{y}_{|0} - \tilde{X}_1 \boldsymbol{\gamma}) = 0$ of the partial regression.

Step 4. $\tilde{X}_1$ has full column rank: if $\tilde{X}_1 \mathbf{c} = 0$ then $X_1 \mathbf{c} = H_0 X_1 \mathbf{c}$ lies in the column space of X_0, which by full rank of X forces $\mathbf{c} = 0$. So the solution is unique, $\hat{\boldsymbol{\gamma}} = \hat{\boldsymbol{\beta}}_1$, and the residual is $\mathbf{y}_{|0} - \tilde{X}_1 \hat{\boldsymbol{\gamma}} = \mathbf{e}$.

(b) The partial regression has residual sum of squares $\mathbf{e}'\mathbf{e} = \mathbf{y}'(I - H)\mathbf{y}$. Under $\boldsymbol{\gamma} = 0$ its residual is $\mathbf{y}_{|0}$ itself, with sum of squares $\mathbf{y}' M_0 \mathbf{y} = \mathbf{y}'(I - H_0)\mathbf{y}$. The difference is $\mathbf{y}'(H - H_0)\mathbf{y}$, so

$$F = \frac{\mathbf{y}'(H - H_0)\mathbf{y}/q}{\mathbf{y}'(I - H)\mathbf{y}/(n - p - 1)},$$

which is (10.1). Software applied naively to the partial regression would use $n - q$ residual degrees of freedom. The correct value is $n - p - 1$, because $\mathbf{e}$ lies in the $(n - p - 1)$-dimensional orthogonal complement of the column space of X. ■

Exercise 10.7 (The Huber location estimator)

Fix $\hat{\tau} > 0$ and $k > 0$, let $\psi(u) = \max(-k, \min(k, u))$, $u_i(\mu) = (x_i - \mu)/\hat{\tau}$ and $g(\mu) = \sum_i \psi(u_i(\mu))$.
(a) Show that the solutions of $g(\mu) = 0$ form a nonempty closed bounded interval. Show that it is a single point unless no observation lies within $k\hat{\tau}$ of the solutions.
(b) Show that every solution $\hat{\mu}$ satisfies $\hat{\mu} = \sum_i w_i x_i / \sum_i w_i$ with $w_i = \min\{1, k/|u_i(\hat{\mu})|\}$ (the fixed-point equation of iteratively reweighted least squares).
(c) Show that if $\max_i |x_i - \bar{x}| \le k\hat{\tau}$, then $\hat{\mu} = \bar{x}$.

Solution [Supplement].

(a)

Step 1. Each term $\psi(u_i(\mu))$ is continuous and nonincreasing in μ, hence so is g. Also $g(\mu) = nk$ for $\mu \le \min_i x_i - k\hat{\tau}$ and $g(\mu) = -nk$ for $\mu \ge \max_i x_i + k\hat{\tau}$.

Step 2. By the intermediate value theorem g has a root. The root set $g^{-1}(\{0\})$ is closed by continuity and bounded by Step 1. If $g(\mu_1) = g(\mu_2) = 0$ with $\mu_1 < \mu_2$, monotonicity forces $g = 0$ on $[\mu_1, \mu_2]$, so the root set is an interval.

Step 3. If $g \equiv 0$ on $[\mu_1, \mu_2]$ with $\mu_1 < \mu_2$, then every term, being nonincreasing with a constant sum, is constant there. But a term with $|u_i(\mu)| < k$ at some $\mu \in (\mu_1, \mu_2)$ is strictly decreasing near μ. So a nondegenerate root interval requires every observation to be at distance at least $k\hat{\tau}$ from every point of (μ_1, μ_2) (with equally many on each side, as $g = 0$). Otherwise the root is unique.

(b) For $u_i \neq 0$ write $\psi(u_i) = w_i u_i$ with $w_i = \psi(u_i)/u_i = \min\{1, k/|u_i|\} \in (0, 1]$. For $u_i = 0$ both sides are 0 with $w_i = 1$. Then $0 = \sum_i w_i (x_i - \hat{\mu})/\hat{\tau}$, and since $\sum_i w_i > 0$, $\hat{\mu} = \sum_i w_i x_i / \sum_i w_i$. Observations beyond $k\hat{\tau}$ get weight proportional to $1/|u_i|$.

(c) At $\mu = \bar{x}$ all $|u_i| \le k$, so $\psi(u_i) = u_i$ and $g(\bar{x}) = \sum_i (x_i - \bar{x})/\hat{\tau} = 0$. Suppose the root set were an interval $[\mu_1, \mu_2] \ni \bar{x}$ with $\mu_1 < \mu_2$. If all $x_i = \bar{x}$, any $\mu \in (\mu_1, \mu_2)$ close to $\bar{x}$ has $|u_i(\mu)| < k$. Otherwise there are observations on both sides of $\bar{x}$, and moving μ from $\bar{x}$ into (μ_1, μ_2) toward one of them gives $|x_i - \mu| < |x_i - \bar{x}| \le k\hat{\tau}$ for that x_i. Either case contradicts Step 3 of (a), so $\hat{\mu} = \bar{x}$. The Huber estimator coincides with the mean when there are no outlying points and departs from it only when some residual exceeds $k\hat{\tau}$. ■